Cost Effective Housing Techniques

Unit I: Housing Scenario, Finance and Planning

Syllabus

a) Housing scenario: status of urban housing, status of rural housing. b) Housing finance: existing finance system in India, government role as facilitator, status of rural housing finance, impediments in housing finance and related issues. c) Land use and physical planning for housing: planning of urban land, Urban Land Ceiling and Regulation Act, efficiency of building bye-laws, residential densities. d) Housing the urban poor: living conditions in slums, approaches and strategies for housing the urban poor.

1. Introduction to Housing

Housing is not just a shelter of four walls and a roof. It includes the **house, its surroundings, infrastructure services** (water, sanitation, power, roads) and **social facilities** (school, health, market) which together give a safe and healthy living environment. After food and clothing, shelter is the third basic need of man.

Importance of housing

- Gives **security, privacy and dignity** to a family; improves **health** and productivity
- Housing is a major part of the **national wealth** (about 20-30% of capital formation)
- The construction sector gives **employment** to millions, second only to agriculture
- Housing has strong **multiplier effect** on 250+ allied industries (cement, steel, bricks, paint, timber)
- Poor housing leads to slums, disease, crime and social unrest

Meaning of cost-effective housing

Cost-effective (low cost) housing means reducing the cost of construction by proper **planning, design, materials and techniques** without sacrificing **strength, durability and comfort**. It is NOT cheap or poor-quality construction.

Ways of reducing cost

* Use of **local materials** and substitutes for scarce materials
* **Improved techniques** (precast, prefabrication, filler slab, arches)
* **Efficient planning**: reduce built-up area, avoid wastage of space, optimum room sizes
* Reduce **wall thickness, use of rat-trap bond, hollow blocks**

* Standardisation, modular coordination and **mass production**
* Cost of land and services reduced through **planned layouts**, higher density

2. Housing Scenario in India

India has a huge **housing shortage** that is both **quantitative** (not enough dwelling units) and **qualitative** (unserviceable kutcha houses, congestion, no services).

Housing shortage (approximate figures)

Item	Figure / Remark
Urban housing shortage (Technical Group, 2012)	About **18.8 million** dwelling units (2012), 95% in EWS and LIG
Rural housing shortage (2012)	About **43.7 million** dwelling units
Slum population (Census 2011)	About **65 million** persons (17% of urban population), 13.9 million slum households
Households without proper latrine / water	Large share (improving under Swachh Bharat, Jal Jeevan)
Distribution of shortage	Mostly **EWS (Economically Weaker Section)** and **LIG (Low Income Group)**

- Figures change with every census / survey; for exams remember the **order of magnitude** and the trend: shortage is mostly among the poor
- Causes: **rapid population growth, urbanisation**, rural-urban migration, high cost of land and building materials, poor housing finance, poor planning and lack of services

3. Status of Urban Housing

- India's urban population has grown from about **28% (2001) to 31% (2011)** and is expected to cross 40% by 2030
- Cities have a **dual housing market**: a formal sector (private builders, public agencies) for rich and middle class, and an **informal sector** (slums, squatters, unauthorised colonies) for the poor
- **Shortage of land and its high cost** is the main problem in metros; affordable housing is pushed to the city edges
- **Slums:** about one in six urban Indians lives in a slum; **congestion** (more than 1 family per room) is high

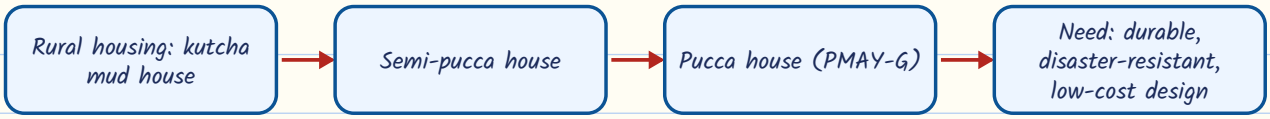
- **Rental housing** has declined because of rent control laws, which discouraged owners from building rental houses
- Government agencies: **State Housing Boards, Development Authorities, HUDCO, municipal corporations**; **PMAY-Urban (2015)** aims at housing for all

Housing categories by income (PMAY norms)

Category	Annual household income	Dwelling size (carpet area)
EWS	Up to Rs 3 lakh	Up to 30 sq.m
LIG	Rs 3 - 6 lakh	Up to 60 sq.m
MIG-I	Rs 6 - 12 lakh	Up to 160 sq.m
MIG-II	Rs 12 - 18 lakh	Up to 200 sq.m

4. Status of Rural Housing

- About **65-70% of the population** lives in villages; rural housing shortage is mainly **qualitative**: most houses are **kutcha or semi-pucca** (mud walls, thatch roofs)
- Types: **kutcha** (mud, bamboo, thatch), **semi-pucca** (brick walls, mud/tiled roofs), **pucca** (burnt brick / stone / concrete with RCC or tiled roof)
- Problems: poor **durability**, damage in floods and cyclones, **no sanitation, smoke from chulhas**, overcrowding, low income and no formal credit
- Rural houses are **self-built** with local labour and local materials; use of **local skills** is high but technology is poor
- Schemes: **Indira Awaas Yojana (IAY, 1985)**, now **PMAY-Gramin (2016)** with a target of 2.95 crore houses; unit assistance **Rs 1.2 lakh (plains), Rs 1.3 lakh (hills / difficult areas)**, house of minimum **25 sq.m** with a hygienic cooking space, support for toilet (SBM) and 90-95 days MGNREGA wage employment



5. Housing Finance

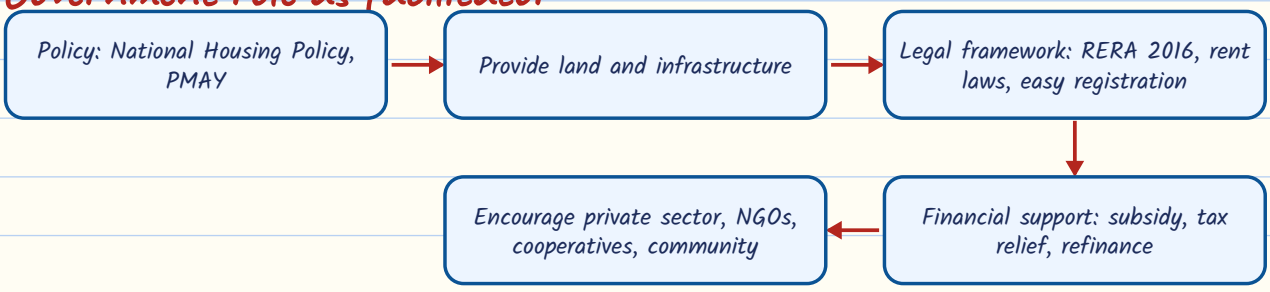
Housing is costly: a house may cost **5-10 times the annual income** of a family. Hence **long-term credit** is needed. **Housing finance** is the system that mobilises funds and lends them for purchase, construction or improvement of houses.

Existing finance system in India

Institution	Role
HUDCO (Housing and Urban Development Corporation, 1970)	Government techno-financial institution; loans to state boards and agencies, **EWS / LIG housing and slum improvement**, rural housing
National Housing Bank (NHB, 1988)	Apex regulator, subsidiary of RBI; refinances housing finance companies, regulates HFCs (now under RBI)
HDFC (1977), LIC Housing Finance, PNB Housing, Can Fin Homes etc.	**Housing Finance Companies (HFCs)** - private, market-based long-term loans
Commercial banks	Major share of housing loans; **priority-sector lending**
Cooperative banks, Housing cooperatives, NBFCs	Local lending, micro-finance
Employers, provident funds, GIC, LIC	Housing loans / advances to employees, funds for housing agencies
State housing boards and Development authorities	Build and allot houses with hire-purchase

- **Sources of funds:** deposits, bonds and debentures, **refinance from NHB**, government equity, **external agencies (World Bank, HUDCO overseas)**, **provident funds, insurance**
- **Government incentives:** income-tax deduction on loan principal and interest (Sections 80C, 24b), **Credit Linked Subsidy Scheme (CLSS)** under PMAY, **priority sector** status, **RBI norms on loan-to-value**

Government role as facilitator



- The government moved from a **direct builder** to an **enabler / facilitator**: it makes **land available, sets standards and bye-laws, gives subsidies and reforms laws**, while the people and the private sector build
- Instruments: **National Urban Housing and Habitat Policy**, **Real Estate (Regulation and Development) Act (RERA, 2016)** to protect buyers, **Affordable Housing Fund**,

interest subsidy, **Credit Risk Guarantee Fund**, **Rental Housing (ARHC)**, **land pooling**, FAR relaxation, single-window clearance

Status of rural housing finance

- Formal credit is **very limited** in villages: low and irregular incomes, no clear **land titles / mortgage**, no proof of income
- Sources: **NABARD** refinance, **Regional Rural Banks**, cooperative banks, **HUDCO** (rural housing loans), micro-finance institutions, **self-help groups (SHGs)**, and traditional money-lenders at high interest
- **Grants** under PMAY-G are the main support; loan up to **Rs 70,000** is permitted for the beneficiary under PMAY-G (subject to rules)
- Need: **micro-housing finance**, incremental lending, community guarantees, and land rights

Impediments in housing finance and related issues

- **Low incomes** of the target group; lack of **repayment capacity**; irregular, informal-sector earnings
- **Unclear land titles**, difficult mortgage and foreclosure laws; high **stamp duty and registration charges**
- High **interest rates**, high **loan-to-value** expectations, **long processing** and documentation, collateral needs
- **Shortage of long-term funds** (short-term deposits financing long-term loans: asset-liability mismatch)
- **Rent control laws**, building bye-laws not matching the poor, **delays in approval**, high cost of land
- **Non-performing assets**, project delays by developers, cost escalation
- Weak role of **cooperatives** and lack of **innovative products** for the informal sector

6. Land Use and Physical Planning for Housing

Land is the most scarce and costly input of housing. In big cities **land cost is 40-60%** of the cost of a house. Proper **land use planning** gives orderly development, efficient services and affordable housing.

Planning of urban land

- **Master Plan / Development Plan** (20 yrs) fixes **land use zones**: residential, commercial, industrial, recreation, public, transport, agriculture
- **Zoning regulations** control the use and density; **Zonal / Local area plans** (5 yrs), and **town planning schemes**
- **Neighbourhood concept (Clarence Perry):** a unit of about **5000 population**, with primary school at centre (walking 400-800 m), shops, park, and no through traffic
- Typical land distribution in a residential area: **residential 45-50%, roads 15-20%, open and recreation 15-20%, community facilities 10-15%**
- **Land policy tools:** acquisition, **land pooling and readjustment**, land banks, **Transfer of Development Rights (TDR)**, reservation of **EWS / LIG plots** (typically 15-25% of land / floor area in new projects), **sites and services** schemes, cross-subsidy
- **Strategies for land supply:** release of **government / surplus land**, redevelopment of old areas, **vertical growth with higher FSI**, optimum use of vacant land

Urban Land (Ceiling and Regulation) Act, 1976

- Passed to **prevent concentration of urban land** in few hands, to **prevent speculation and profiteering**, and to make **land available for housing the weaker sections**
- Fixed a **ceiling limit** on vacant land that a person could hold in an urban agglomeration (**500 to 2000 sq.m** depending on the city category); the **excess land** was to be taken over by the state and used for housing for the poor
- **Failures:** litigation, delays, **land hoarding and black market** instead of supply; **fall in land supply**; escape by legal loopholes; the Act did **not** produce housing for the poor
- **Repeal:** the **Urban Land (Ceiling and Regulation) Repeal Act, 1999** repealed the Act (adopted by states); land prices and supply improved afterwards in many places
- Present approach: **encourage land development by the private sector, regulation of land use**, not ceiling

Efficiency of building bye-laws

Building bye-laws are rules made by the local body to regulate construction for **safety, health, convenience** and orderly growth.

Provision	Meaning
FAR / FSI (Floor Area Ratio)	Total floor area / plot area; e.g., FAR 2.0 means built-up area twice the plot

Provision	Meaning
Ground coverage	Percentage of the plot covered by the building (30-60%)
Setbacks	Front, rear, side open spaces for light, ventilation and fire safety
Height of building	Limited by street width, airport rules, fire norms
Minimum room sizes	Habitable room 9.5 sq.m (NBC); kitchen 4.5 sq.m; bathroom 1.8 sq.m
Plot size	Minimum plot size for EWS reduced to 20-30 sq.m
Parking, staircase, lift, fire exits	Safety and services

- **Problems with existing bye-laws:** based on old standards, **too rigid and costly** for the poor, hinder **incremental housing**, high set-back / plot-size requirements, multiple clearances and delays
- **Reforms: model bye-laws, NBC 2016** compliance, **relaxed standards for EWS / LIG** (smaller plots and rooms, row housing, mixed use), **single-window, online approval**, higher **FSI near transit**, **performance-based** rules

Residential densities

Density = number of persons (or dwelling units) per hectare. It controls the **cost of land and services per family**.

Types of density

* **Net residential density** = population / area of residential plots only (excluding roads, parks)
* **Gross density** = population / total area including roads, open spaces, facilities
* **Neighbourhood density** and **city density**
* Dwelling units per hectare (**DU/ha**), **habitable rooms** per hectare, **FAR** / built-up area

Density type	Persons per hectare (approx.)
Low density	Less than 75 (bungalows)
Medium density	75 - 200
High density	200 - 500 (flats, row houses)
Very high (slums / congested)	More than 500

- **Higher density reduces the per-family cost** of land, roads, water, sewer lines, and electrification; typical Indian norms for planned cities are **100 - 300 persons / ha** (and plotted EWS schemes 300-500)
- Too high density: **congestion, lack of light and air, health, traffic, poor services**; so balance is needed

- **Plot size and density:** a 30 sq.m EWS plot gives about 250-300 DU/ha; 60 sq.m LIG plot about 100-125 DU/ha; multi-storey flats reach 300-600 persons/ha with open space left

7. Housing the Urban Poor

The urban poor (EWS and LIG) cannot afford formal housing: land, building cost and loan repayments are beyond their income. They live in **slums, squatter settlements, footpaths, rented rooms**.

What is a slum?

- As per the **Slum Areas (Improvement and Clearance) Act, 1956**, a slum is an area unfit for human habitation because of **dilapidation, overcrowding, faulty design, lack of ventilation, light, sanitation** or any combination of these
- Census 2011 definition: compact area of at least **300 population or 60-70 households** of poorly built, congested tenements in unhygienic conditions with inadequate infrastructure
- Types: **notified, recognised and identified slums**; squatter settlements; **pavement dwellers**; unauthorised colonies

Living conditions in slums

Aspect	Condition
Housing	Temporary / kutcha structures of plastic, tin, bamboo, mud; **overcrowded** (5-6 persons in a 10 sq.m room)
Water supply	Public taps / hand pumps, shared by many; irregular and unsafe
Sanitation	No toilets, **open defecation**, open drains, garbage dumped
Environment	Narrow lanes, no light and ventilation, flood prone low lands, near rail tracks, nalas
Health	High incidence of **TB, diarrhoea, malaria, dengue, skin diseases**; high infant mortality
Social / economic	Low and irregular income (informal work), **insecurity of tenure**, no credit, crime, fire hazards, poor education
Services	Few schools, clinics, no electricity or illegal connections

Approaches and strategies for housing the urban poor

Approach	Idea
Slum clearance and redevelopment	Demolish and rebuild (old approach); causes displacement and loss of livelihood, costly
In-situ slum upgradation	Improve **services and houses at the same place**: water, drains, toilets, lanes, security of tenure; minimal displacement
Slum rehabilitation / redevelopment with private participation	Land of slum used as resource; **free-sale component** cross-subsidises the free houses of slum dwellers (Mumbai SRA, **Slum Rehabilitation Scheme**)
Sites and services	Government provides **plots with water, sewer, roads** and the poor build their house **incrementally** with loans / own labour; cheap and flexible
Environmental Improvement of Urban Slums (EIUS)	Basic services, central sector (1972)
Relocation / resettlement	Shifted to the periphery; fails if far from jobs
Rental and dormitory housing	For migrants; **Affordable Rental Housing Complexes (ARHC)**
Community participation: NGOs, CBOs, SHGs, cooperatives	Beneficiary-led construction, savings groups

Government programmes (urban poor)

* **VAMBAY (2001)** Valmiki Ambedkar Awas Yojana; **NSDP**; **JNNURM – BSUP and IHSDP (2005)**
* **Rajiv Awas Yojana (RAY, 2011)**: slum-free India, **property rights** to slum dwellers
* **PMAY-Urban (2015)** four verticals: **In-situ Slum Redevelopment (ISSR)**, **Credit Linked Subsidy Scheme (CLSS)**, **Affordable Housing in Partnership (AHP)**, **Beneficiary-Led Construction (BLC)**
* **Basic Services to the Urban Poor (BSUP)**, Swachh Bharat, Smart Cities, **DAY-NULM** (livelihoods)

- **Strategies:** security of tenure **(pattas)**, incremental housing, convergence of schemes (water, sanitation, livelihood), participation, **low-cost technology and local materials**, planned land reservation (15-25%), **mixed income housing**, rental options, linkage to **employment**

8. National Housing Policies and Programmes

Policy / Programme	Key points
National Housing Policy 1988	First policy; target of **housing for all**, special focus on the poor, creation of **NHB**, shift to facilitator role
National Housing and Habitat Policy 1998	Public-private partnership, **cooperatives**, rural housing, use of **cost-effective technology**, role of **Building Centres**
National Urban Housing and Habitat Policy 2007	**Affordable housing for all**, 10-15% land or 20-25% FAR for EWS/LIG in new projects, **PPP**, special economic zones
Jawaharlal Nehru National Urban Renewal Mission (2005)	BSUP and IHSDP for the urban poor
PMAY (Urban and Gramin)	Housing for All by 2022 (extended)
Affordable Housing Fund (NHB, 2018)	Funding for the affordable housing, refinance of HFCs
RERA 2016	Registration of projects and agents, consumer protection

Building Centres

- Set up by **HUDCO** and **NGOs** from 1988 to **spread cost-effective technology** to the common man
- Functions: **production of building components** (blocks, tiles, doors), **training of artisans and masons**, **demonstration** of techniques, technical advice, **material banks**
- Examples: Nirmithi Kendras (Kerala), Building Centres, **CBRI (Roorkee)**, **SERC (Chennai)**, **BMTPC (Delhi)** - research and promotion of technology

9. Cost Components of a House and Planning for Economy

Component	Share of total cost (approx.)
Land	10 - 40% (depends on city)
Building materials	55 - 65% of construction cost
Labour	15 - 25%
Services (water, power, roads)	10 - 15%
Fees, taxes, finance charges	5 - 10%
Profit / overheads	5 - 10%

- **Walls 25-30%, roof and floors about 20%, foundations 10-15%, finishes 15-20%** - so savings in walls and roof give biggest gain
- **Planning for economy:** compact plans, **reduce circulation area**, avoid long passages, group the wet areas (kitchen-toilet) with common plumbing wall, use of **row housing / semi-detached** (common walls), standard **modular dimensions (multiples of 100 mm)**, simple elevations without projections, adequate **ventilation**
- Optimum **building height** (3-4 storeys walk-up without lift for EWS/LIG) - cuts cost of lifts and structure
- **Plinth area** to carpet area ratio about 0.8; plinth area rates are used for estimation

Layout types for low cost housing

Layout	Features
Row housing (terrace)	Houses share side walls, **less land and boundary**, cheap services; used for EWS/LIG; 15-20 houses per row max
Cluster housing	Houses around a common open court, **safe play and community space**, shorter road and service lengths
Detached / semi-detached	More land; costly
Walk-up flats (G+3 / G+4)	High density, shared staircase; **low cost of land per family**
Plotted development with sites and services	Plots 20-60 sq.m, incremental growth

Principles of site selection for housing

* **Accessibility** to work place and transport, **availability of services**
* Good **drainage**, no flooding, firm sub-soil, away from industrial pollution
* **Cost of land** and legal title clear; room for future expansion
* Near **schools, health and market**; avoid agricultural or forest land

10. Quick Revision Points

Remember for exam

* Housing shortage: urban about 18.8 million, rural about 43.7 million (2012); mostly EWS and LIG
* Government role has changed from builder to **facilitator / enabler**
* HUDCO 1970, NHB 1988, HDFC 1977; RERA 2016; PMAY-U 2015, PMAY-G 2016
* ULCRA 1976 repealed in 1999 because it failed to deliver land
* Density = persons per hectare; higher density = lower cost of services
* In-situ upgradation, sites and services and slum rehabilitation are the key approaches for the poor

Important questions

1. Explain the housing scenario in urban and rural India. What are the causes of housing shortage?
2. Describe the existing housing finance system in India and the role of government as a facilitator.
3. Discuss the impediments in housing finance and the status of rural housing finance.
4. Explain the planning of urban land. What are the objectives and drawbacks of the Urban Land Ceiling Act?
5. Discuss the efficiency of building bye-laws and suggest reforms for low cost housing.
6. Define residential density and explain its types and effects on cost.
7. Describe the living conditions in slums.
8. Explain the approaches and strategies for housing the urban poor.